Personal AI for Continuous Spatial Collaboration

Communication and Perception for Situated Action

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Abstract

I propose to investigate personal AI that can participate in ongoing activities: understanding speech in relation to what a person sees and does, comparing alternatives, and translating discussion into actions in software and spatial environments. Spatial design across desktop and extended reality (XR) will provide an initial setting for studying this broader vision. The central mechanism is a jointly revisable task representation connecting expressions, object references, goals, design rationales, constraints, and action outcomes. Through formative research, prototype development, controlled studies, and a multiweek deployment, I will examine how this representation supports evolving intentions, effective execution, and continuity across sessions. Expected contributions include an empirical account of continuous human-AI collaboration, interaction mechanisms that connect shared understanding with action, and evaluation methods covering task outcomes, repair costs, and sustained use. Personal preferences and a sense of familiarity will be investigated as distinct, exploratory dimensions of the collaboration.

1 Motivation and Research Context

My interest in personal AI concerns the possibility of working with an intelligent collaborator throughout an activity. People develop intentions while talking, pointing, trying alternatives, consulting information, and revising earlier decisions. They also leave tasks unfinished and return through another device. Supporting this process requires an AI to connect conversation, situated observations, and actual changes to the work.

Consider someone redesigning a workspace: "Make this area suitable for two people to collaborate, but keep a workstation by the window." The AI uses the current view, pointing, and scene objects to propose layouts the user can experience. The user then moves a desk manually and narrows the request to one group of furniture. The AI must interpret the revised scope, compare feasible arrangements, and implement a useful local change. Returning the next day, the user should be able to recover the reasoning behind the layout and continue developing it.

Here, personal means adapting to a person's tasks, shared working history, and revisable preferences. Spatial design makes these capabilities observable because language, viewpoint, direct manipulation, and measurable outcomes interact. I will begin with a bounded asset library and resettable virtual rooms, connecting information comparison, scene editing, and explanation. XR provides situated input and spatial feedback; it is a research context within the broader goal of personal AI across activities.

2 Related Research and Positioning

From speech to actionable intentions

Rambler supports speech-based writing through semantic operations on ideas [5], while Orality externalizes spoken thinking into a manipulable semantic canvas [3]. These systems motivate studying intentions as something developed through expression and interaction. Rambler in the Wild identifies needs for contextual and personalized conversational support [6], and IntentPrism makes inferred intentions inspectable and editable [7]. I aim to extend this perspective to situations where revising an idea must also change references to spatial objects and the actions performed on them.

Connecting context with action

Reference-Based Manipulation organizes spatial references through Source, Anchor, and Frame, including references to historical states [1]. EchoLadder connects abstract design requests with applicable, reversible

scene suggestions [2]. PANDALens links contextual capture, spoken contributions, and situated writing [4]. Together, these works provide foundations for linking communication, perception, and execution. They also motivate investigating how these links evolve when user actions and AI processing interleave.

Related work already addresses multimodal action prediction and interactive world generation [11,12]. MUSE combines requirement management, scene memory, local execution, and verification, including controlled multiround editing [13]; SpatiaLLM uses constraint-validated spatial memory for MR content placement [14]. My proposed contribution concerns how users and AI jointly revise the meaning of an ongoing task, and how that process affects collaboration and resumption. MUSE's reconstruction of working memory each turn further motivates examining which goals and rationales should persist, and how local edits can be distinguished from changes to higher-level intentions.

Evaluating capability and experience

InterFlow highlights the interaction between assistance and an ongoing workflow, while also revealing the need to assess technical correctness separately [8]. Desirable Unfamiliarity distinguishes visual demands during speaking and reviewing [9], and Sensible Agent investigates context-dependent assistance in AR [15]. Personal affective physicalization foregrounds users' interpretation and revision of personal meaning [10]. These studies inform an evaluation that separates task performance, collaboration processes, and subjective experience.

3 Research Questions

The central question is how personal AI and its user can maintain a shared, revisable understanding of an activity as communication, context, and action outcomes change.

RQ1 Shared Understanding during Expression and Manipulation

When speech, pointing, gaze, and manual edits provide incomplete or conflicting information, how can the system identify objects, scope, and unresolved intentions? I will investigate object-linked expressions and directly revisable task representations. The expectation is reduced repetition and scope errors, with benefits depending on task complexity and the effort required to maintain the representation.

RQ2 Situated Reasoning and Effective Action

How can the AI compare feasible alternatives and execute them while the user continues editing or changing requirements? I will examine linking actions to the objects, constraints, and task versions on which they depend, assessing completion, design quality, repair costs, and the latency introduced by verification.

RQ3 Continuity across Sessions and Devices

Which evidence, decisions, rejected alternatives, and preferences should persist when users switch devices or return later? I will examine inspectable and revisable task memory, focusing on resumption, stale information, and actual adoption. Familiarity and relational experience will be explored separately.

4 Proposed System and Interaction Mechanisms

4.1 Grounding communication in the current activity

The prototype will combine speech, pointing or selection, viewpoint, and scene events. Initial studies will use reliable object identifiers and editor state; first-person imagery and eye tracking can be introduced subsequently. Timestamped inputs will distinguish the scene at speaking time from the scene at execution time. Gaze will provide evidence about candidate objects alongside language and action. The interface will support continuing speech, references to earlier objects, and local revisions during manipulation.

4.2 Making task understanding jointly revisable

A task representation will link goals, objects, constraints, rationales, and actions, recording each item's source, scope, time, and status. It will distinguish explicit requests, AI interpretations, alternatives under

consideration, and superseded decisions. Users can revise entries through speech or direct manipulation, with changes propagated to dependent plans. For example, keeping a workstation by the window describes a purpose and spatial relationship rather than necessarily fixing furniture coordinates. A manual move may revise a solution, test an alternative, or change the goal. Local previews and focused clarification will help establish which interpretation applies.

4.3 Connecting judgments to verifiable execution

Language and vision models will interpret requests, retrieve information, and propose alternatives. Explicit tools will query objects, calculate geometry, update scenes, and check predefined conditions. The initial toolset will cover a fixed information collection, object selection and transformation, layout comparison, and explanations tied to implemented changes. The AI can also contribute judgments, such as explaining why one layout may better support conversation and enabling users to experience alternatives. Actions will retain their dependencies and scene versions; relevant changes will trigger local recomputation. Evaluation will distinguish a model's claim of success, successful tool execution, and satisfaction of the user's goal. Open-ended qualities such as comfort will require user judgment and comparative assessment.

4.4 Continuing a shared working history

On resumption, the system will expose active goals, key decisions, and their evidence for revision or deletion. Temporary choices will remain distinct from preferences inferred across tasks. Feedback may take the form of speech, object-linked annotations, spatial alternatives, or detailed explanations. Repeated use will inform the study of stable communication styles and personal meaning.

5 Research Methodology and Evaluation

Study 1 Understanding continuous collaboration

A formative study with approximately 12-16 participants of varied spatial design experience will examine ideation, editing, and task resumption. Contextual interviews, interaction recordings, and selected Wizard-of-Oz sessions will reveal how spoken revisions, references, direct manipulation, and rationales combine. Human simulation will identify interaction requirements; implemented model capabilities will be evaluated separately. Outcomes will include a classification of coordination and breakdowns, task units, and representative events for subsequent studies.

Study 2 Separating representation and interface effects

Technical evaluation will use event sequences covering ambiguous references, local revisions, concurrent manual edits, new constraints, and superseded decisions. Independent annotation will identify acceptable outcomes for specified tasks, retaining multiple valid interpretations where appropriate. Development and test sets will be separated by user and scene.

Comparisons will include conversation plus the current scene, a strong baseline using retrieved event logs or structured summaries, and the proposed representation with updated dependencies. Conditions will share model versions, tools, and available raw information, with matched context and computation budgets and recorded costs. Two separate comparisons will first hold the interface constant to assess state mechanisms, then hold the backend constant to assess directly editable, object-linked feedback. Adaptations of mechanisms such as MUSE will document implementation differences.

Controlled studies will initially be planned around 24-36 participants, with final recruitment informed by pilots and power analysis. Counterbalanced within-participant designs will include specified-constraint and open-ended tasks. Model configurations will remain fixed, and naturally occurring events will be distinguished from experimentally introduced changes.

Question	Primary measures	Complementary evidence
RQ1	Object and scope errors; repair turns	Revision strategies; retrospective accounts
RQ2	End-to-end completion; active constraint satisfaction	Unrelated changes; time and repair cost; design quality
RQ3	Resumption time; stale-information errors	Return use; adaptation; familiarity

Completion will be measured across the full task episode, including timeouts, non-execution, and partial completion. Updated goals will define success when users explicitly change them. Specified tasks will use their task descriptions as evaluation criteria; open-ended tasks will use participant confirmations and event reviews. Condition-blind assessors will judge whether outputs meet those requirements, with agreement reported. Analyses will account for repeated observations and participant and task variation, reporting effect sizes and confidence intervals with prespecified primary comparisons. Interviews and workload measures will help explain outcomes.

Study 3 Examining continuity and sustained use

A controlled next-day resumption study will compare session summaries with task memory preserving sources and status. Conditions will match delays and device transitions, with counterbalanced tasks and order. An exploratory deployment of a stable version will then involve approximately 12-18 participants over 4-6 weeks on small real design projects. This deployment will examine adoption and mechanisms in use, without making causal claims from crossover comparisons. Logs and repeated interviews will investigate useful and misapplied history, revisions, completed workflows, familiarity, and communication preferences, considering novelty and actual reliability.

6 Contributions and Research Schedule

Expected contributions are: an empirical account of how intentions develop through speech and manipulation; evaluated mechanisms connecting shared task representations with situated action; and tasks and methods for assessing understanding, execution, repair, and continuity. Subject to consent and licensing, I will release prototypes and de-identified event sequences or synthetic evaluation cases.

Stage	Main activities	Expected outputs
Year 1	Formative research; desktop/XR prototype	Behavioral account; scoped tasks
Year 2	Representation and execution mechanisms; ablations	Working system; technical evidence
Year 3	Controlled studies; cross-device resumption	Interaction and continuity findings
Year 4	Multiweek deployment; synthesis and writing	Design knowledge; doctoral thesis

This four-year plan will be adjusted to the programme and findings. Existing models, reliable scene state, and bounded tools keep the initial scope feasible. Perceptual uncertainty will be evaluated separately before expanding the environment. Studies will obtain informed consent, limit data collection, and support inspection, deletion, and withdrawal. Resettable scenes and local undo will support recovery and analysis.

7 Fit with Can Liu and ERFI Lab

I hope to develop this research with Professor Can Liu because it connects speech-centric intelligent interfaces with embodied interaction in spatial computing. Orality and Rambler motivate studying how intentions emerge through expression; Reference-Based Manipulation and EchoLadder ground the connection to spatial references and executable alternatives; PANDALens informs continuity between situated experience and meaningful output. I am particularly interested in exploring how these directions can meet in a personal collaborator that communicates, perceives, reasons, and acts. Spatial design provides a

tractable starting point for investigating that vision. I would welcome refining the tasks and priorities together in relation to the laboratory's ongoing research.

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